Project: Worked examples
Title: Lifting lug and pin joint
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

A pinned lug between two cheek plates under a load along the lug: tension on the net section through the hole, bearing of the pin on the hole, shear tear-out beyond the hole, and the pin in double shear and in bending, each against an allowable stress; the largest utilisation governs.
$\sigma_t = P/((W - d_h)\cdot t)$; $\sigma_{br} = P/(d_p\cdot t)$; $\tau_{to} = P/(2\cdot a\cdot t)$; $\tau_p = P/(2\cdot\pi\cdot d_p^2/4)$; $M = (P/2)\cdot(t/4 + g + t_s/2)$, $\sigma_b = 32\cdot M/(\pi\cdot d_p^3)$. Each utilisation is the stress over its allowable. The allowables come from the material and the design category in force; the sheet applies no design factor of its own.
Ref: ASME BTH-1 Ch. 3 — Pinned connections
Ref: Shigley's Mechanical Engineering Design Ch. 3 — Shear, bearing and bending stresses

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
P	Load on the lug (along the lug)	120	kN	spec: LOAD-SCHEDULE-01	0.01 … 100000 kN
t	Lug thickness	15	mm	drawing: DRG-LL-102 rev A	1 … 500 mm
W	Lug width at the hole	90	mm	drawing: DRG-LL-102 rev A	5 … 2000 mm
d_h	Hole diameter	26	mm	drawing: DRG-LL-102 rev A	2 … 1000 mm
a	Edge distance, hole edge to lug end (load direction)	30	mm	drawing: DRG-LL-102 rev A	1 … 2000 mm
d_p	Pin diameter	25	mm	drawing: DRG-LL-102 rev A	2 … 1000 mm
t_s	Cheek plate thickness (each)	10	mm	drawing: DRG-LL-102 rev A	1 … 500 mm
g	Clearance, lug face to cheek plate (each side)	1	mm	drawing: DRG-LL-102 rev A	0 … 100 mm
f_t	Allowable tensile stress, lug	150	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa
f_br	Allowable bearing stress, lug	200	MPa	spec: SPEC-MECH-03 design criteria	1 … 3000 MPa
f_v	Allowable shear stress, lug	90	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa
f_b_pin	Allowable bending stress, pin	250	MPa	spec: SPEC-MECH-03 design criteria	1 … 3000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
f_v_pin	Allowable shear stress, pin	150	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa

3. Assumptions

A1. The load acts along the lug's axis through the hole centre. (No out-of-plane or side load.)
A2. The pin bears uniformly over the lug thickness and over each cheek plate; each cheek plate reacts half the load. (Pin-bending idealisation for a lug between two cheek plates.)
A3. A close-fitting pin (d_p ≥ 0.9·d_h). (Uniform-bearing idealisation.)

4. Working

[14] k_fit = d_p / d_h
= 25 mm / 26 mm
= 0.96154
Pin-to-hole ratio

[15] A_net = (W - d_h) * t
= (90 mm - 26 mm) × 15 mm
= 960 mm²
Net section through the hole

[16] sigma_t = P / A_net
= 120 kN / 960 mm²
= 125 MPa
Net-section tension
ASME BTH-1 Ch. 3

[17] A_br = d_p * t
= 25 mm × 15 mm
= 375 mm²
Bearing area

[18] sigma_br = P / A_br
= 120 kN / 375 mm²
= 320 MPa
Bearing of the pin on the hole

[19] A_to = 2 * a * t
= 2 × 30 mm × 15 mm
= 900 mm²
Tear-out area (two planes beyond the hole)

[20] tau_to = P / A_to
= 120 kN / 900 mm²
= 133.33 MPa
Tear-out shear

[21] A_p = 2 * pi * d_p^2 / 4
= 2 × π × (25 mm)^2 / 4
= 981.75 mm²
Pin area in double shear

[22] tau_p = P / A_p
= 120 kN / 981.75 mm²
= 122.23 MPa
Pin shear

[23] M_p = P / 2 * (t / 4 + g + t_s / 2)
= 120 kN / 2 × (15 mm / 4 + 1 mm + 10 mm / 2)
= 585 N·m
Pin bending moment (uniform bearing on lug and cheek plates)

[24] Z_p = pi * d_p^3 / 32
= π × (25 mm)^3 / 32
= 1534 mm³
Pin section modulus
[25] sigma_b = M_p / Z_p
= 585 N·m / 1534 mm³
= 381.36 MPa
Pin bending stress
[26] u_t = sigma_t / f_t
= 125 MPa / 150 MPa
= 0.83333
Net-section utilisation
[27] u_br = sigma_br / f_br
= 320 MPa / 200 MPa
= 1.6
Bearing utilisation
[28] u_to = tau_to / f_v
= 133.33 MPa / 90 MPa
= 1.4815
Tear-out utilisation
[29] u_p = tau_p / f_v_pin
= 122.23 MPa / 150 MPa
= 0.81487
Pin shear utilisation
[30] u_b = sigma_b / f_b_pin
= 381.36 MPa / 250 MPa
= 1.5254
Pin bending utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
sigma_t	125 MPa	Net-section tension
sigma_br	320 MPa	Bearing stress
tau_to	133.33 MPa	Tear-out shear
tau_p	122.23 MPa	Pin shear
M_p	585 N·m	Pin bending moment
sigma_b	381.36 MPa	Pin bending stress
u_t	0.83333	Net-section utilisation
u_br	1.6	Bearing utilisation
u_to	1.4815	Tear-out utilisation
u_p	0.81487	Pin shear utilisation
u_b	1.5254	Pin bending utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Bearing	u_br = 1.6	u_br ≤ 1	1.6	ASME BTH-1 Ch. 3

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Pin bending	u_b = 1.5254	u_b ≤ 1	1.53	
FAIL	Tear-out	u_to = 1.4815	u_to ≤ 1	1.48	ASME BTH-1 Ch. 3
PASS	Net-section tension	u_t = 0.83333	u_t ≤ 1	0.833	ASME BTH-1 Ch. 3
PASS	Pin double shear	u_p = 0.81487	u_p ≤ 1	0.815	

7. Applicability

No warnings.

8. Verification

Template mech.lifting-lug@1.0.0 Content hash 66e02c833f45b372 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0